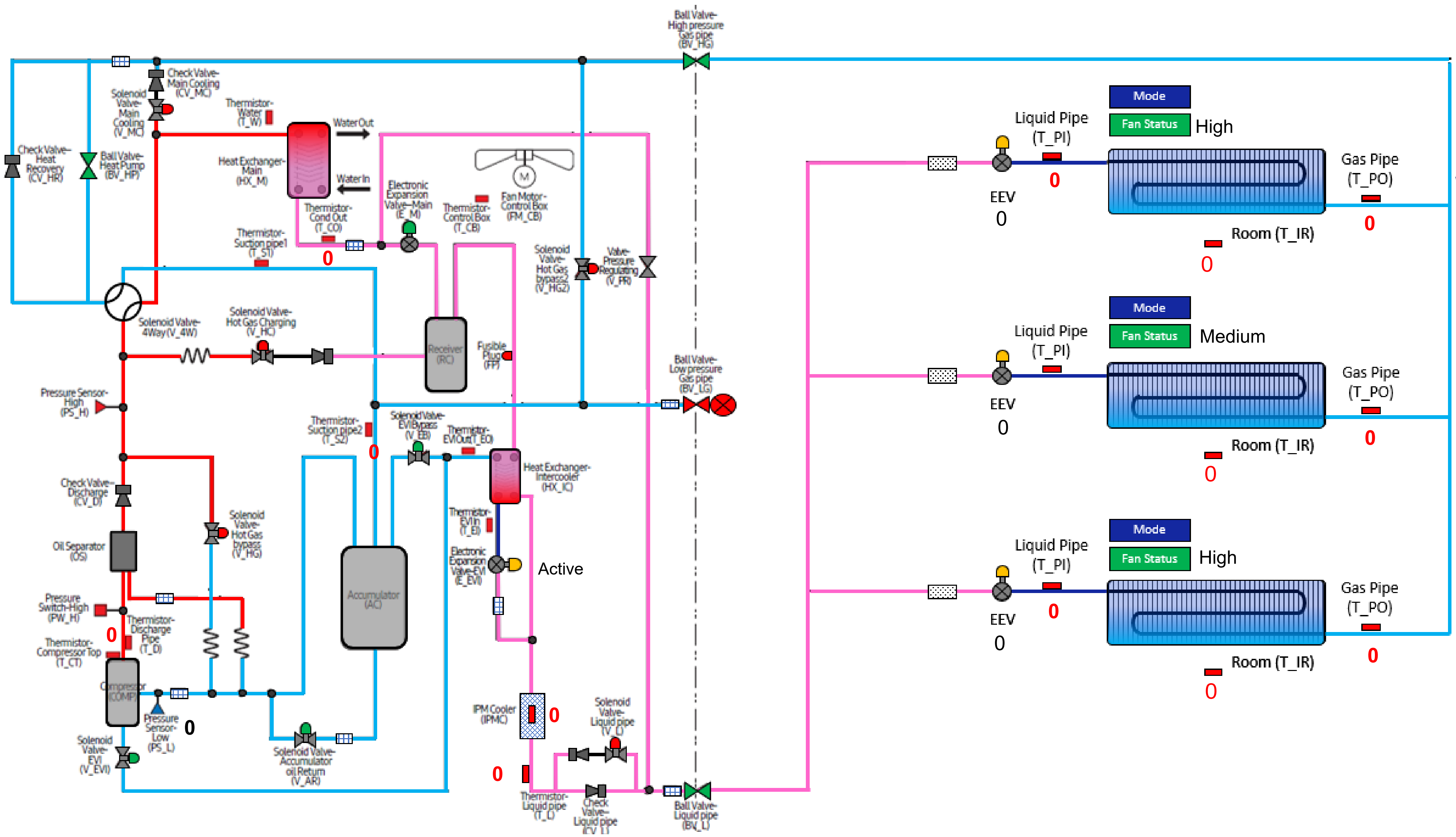
DVM S 3 Phase Water

Heat Pump Operation Evaluation

Cooling Mode



Comments:

Outdoor Unit			
Component	**Target**	**Reading**	
Water Out Temp.			
Compressor Hz.			
High pressure	Based on water temp		Ok
H.P Sat. Temp		0	
Discharge Temp.	150^{0}-212^{0}	0	Ok
IPM Temp,	Below 194^{0}	0	
Low pressure	114 Psig.	0	Low
L.P. Sat. Temp	38^{0}		
Suction 1 Temp.			Ok
Suction 2 Temp		0	
S.H.	3.6^{0} or 9^{0}	*0*	Ok
Min EEV Position	2000 steps		Ok
Cond. Out Temp.	Water Out – (2^{0} – 35^{0})	0	Ok
Liquid Line Temp.		0	Ok
Sub-Cooling	9^{0} or 36^{0}	*0*	Ok

Indoor Units			
Component	**Target**	**Reading**	
IDU 1 EEV		0	Ok
IDU 1 SH	3.6^{0} (Typically 0^{0} – 7^{0})	*0*	Ok
IDU 1 T PI		0	Ok
IDU 1 T PO		0	Ok
IDU 2 EEV		0	Ok
IDU 2 SH	3.6^{0} (Typically 0^{0} – 7^{0})	*0*	Ok
IDU 2 T PI			Ok
IDU 2 T PO		0	Ok
IDU 3 EEV		0	Ok
IDU 3 SH	3.6^{0} (Typically 0^{0} – 7^{0})	*0*	Ok
IDU 3 T PI		0	Ok
IDU 3 T PO		0	Ok

Directions:

Enter the data in the spaces provided. The reading will populate in the refrigerant diagram. Space temperature is entered at the IDU. Use the drop down boxes to identify the state of the component. Use the operating data and the target operation data on the next page to troubleshoot the system.

DVM S 3 Phase Water

Heat Pump Target Operation

Water Source HP Outdoor Unit

Component	Abbreviation		Cooling Mode	
Compressor	Comp	Operation	Target low pressure control (default: 113.8psig)	Primary function, Maintain Average Evaporator Temperature (Average temperature of Eva. In Sensors)
		Range	14~120Hz	Low pressure range 100psig - 128 psig. Evaporator temperature range 45°F -48°F
High side pressure	HP	Operation	Head pressure is effected by the water temperature	Is there a variable flow control valve installed? This will effect head pressure.
		Range	299 psig - 469 psig (water temperature 68°F-104°F)	At 228psig, low limit control is activated at 512psig high limit control is activated
Main EEV	E_M	Operation	Full Open	
		Range	2000 Steps	
EVI EEV	E_EVI	Operation	*Subcooling control (default: SC=36°F)	SC = High Side Saturation Temp. - Liquid Line Temp.
		Range	0-480step	
4 Way Valve	V_4W	Operation	Off	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_W > 104°F or 50Hz↓	Activates Sub-Cooling Control
			Off (Close): T_W < 100.4°F↓ and 65 Hz↑	Discharge Temperature Protection, Control by unit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open when compressor is running)	**Off** during start up if DSH is < 18°F. **Off** for 2 min. if DSH is < 18°F **On** when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	Opens after 20 minutes of continuous run time in cooling mode
Main EEV Solenoid Valve (HR Only)	V_ME	Operation	On (Open)	Energized in cooling only
Main Cooling Solenoid	VMC	Operation	Off (Closed)	
Liquid Pipe Thermistor	T_L	Operation	Used to calculate sub-cooling	Sub-Cooling range is 9°F-36°F
HP - HR Ball Valve (3 phase only)	BV_HP	Operation	Open	Valve must be open for heat pump applications

IDU

Component	Abbreviation		Cooling Mode	Comments
Room Temperature	Room Temp.	Operation	65~86°F	
Liquid Pipe Thermistor	EVA In 1	Operation	Evaporator Saturation Temperature. Range 45-48°F	Used to calculate Evaporator Superheat
Gas Pipe Thermistor	EVA Out 1	Operation	Measures Suction Pipe Temperature	Used to calculate Evaporator Superheat
EEV	EEV 1	Operation	Modulates to maintain Superheat of 3.6 degrees F	Superheat = Pipe out Temp - Pipe in Temp. (Typical Range 0-7)
		Range	0 - 2000 Steps (0 Steps when zone is satisfied)	Typical range between 250 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
		Operation	Low Speed: $\lvert Ts - Ti \rvert \le 1.8$°F	TS - Set Temperature
			Medium Speed: 1.8°F < $\lvert Ts - Ti \rvert \le 5.4$°F	Ti Indoor Temperature
			High Speed 5.4°F < $\lvert Ts - Ti \rvert$	

Ver. 06142023